WESTCLIFF UNIVERSITY | DCS 404 FINAL PROJECT

Machine Learning & Transformer-Based Spam Message Detection

End-to-End NLP Pipeline: SMOTE Imbalance Remediation, Traditional ML & Transformers

Presenter: Sujal | Course: DCS 404 - Artificial Intelligence & Machine Learning | Fall 2026

Featuring: TF-IDF • SMOTE • Logistic Regression • Naive Bayes • DistilBERT • DistilRoBERTa • Streamlit UI

Key Learning #1: Data Hygiene, Anti-Leakage & SMOTE Balancing

Dataset Foundation: UCI SMS Spam Collection corpus containing 5,572 labeled records (ham / spam).

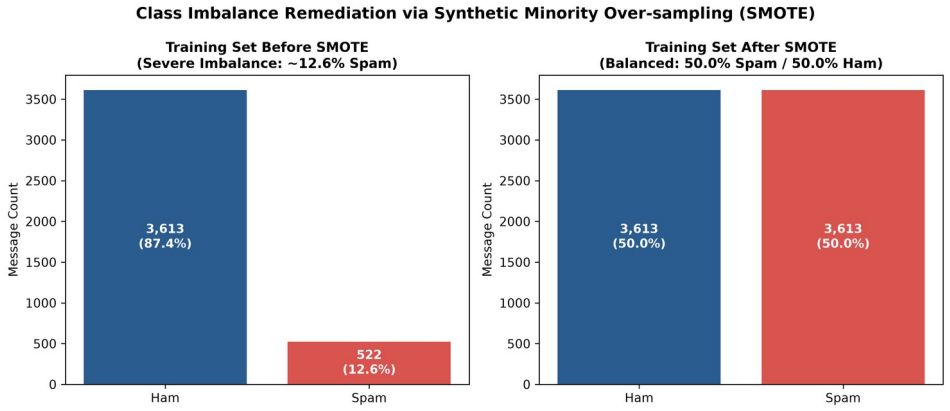
Data Cleaning & Hygiene: Checked missing values (0 nulls). Pruned 403 exact duplicates to yield 5,169 distinct messages (4,516 Ham [87.4%], 653 Spam [12.6%]).

Anti-Data Leakage Protocol: Split dataset strictly first using stratified 80/20 train/test splitting (4,135 train / 1,034 test). The test partition preserves authentic class prevalence.

TF-IDF Vectorization: Fitted TfidfVectorizer (5,000 max features, unigrams + bigrams, English stop words) strictly on training text only before transforming test data.

SMOTE Applied Only to Train: Applied Synthetic Minority Over-sampling Technique (SMOTE) strictly to training TF-IDF features, balancing 3,613 Ham and 522 Spam to an equal 3,613:3,613 distribution.

Test Set Preservation: The 1,034 test messages were never oversampled or modified, ensuring real-world, unbiased empirical evaluation.



Exploratory Data Analysis: Lexical Patterns & Word Clouds

Lexical Explainability: Word clouds provide immediate visual explainability, revealing distinguishing linguistic characteristics between unsolicited and legitimate messages.

Spam Keyword Triggers: Dominated by commercial incentives, urgent calls-to-action, and prizes: 'FREE', 'call', 'claim', 'prize', 'txt', 'won', 'cash', 'urgent', 'customer', 'reply'.

Ham Conversational Vocabulary: Reflects everyday personal communication, social coordination, and colloquialisms: 'come', 'good', 'time', 'got', 'will', 'day', 'ok', 'home', 'later', 'love'.

Message Length Disparity: Spam messages average 138.8 characters (median 149), constrained near the 160 SMS limit due to disclaimers. Ham messages average 71.0 characters (median 52).

Engineering Takeaway: Token frequency disparity confirms that TF-IDF n-grams provide strong discriminative power for linear and probabilistic classifiers.



Key Learning #2: Traditional Machine Learning & SMOTE Trade-Offs

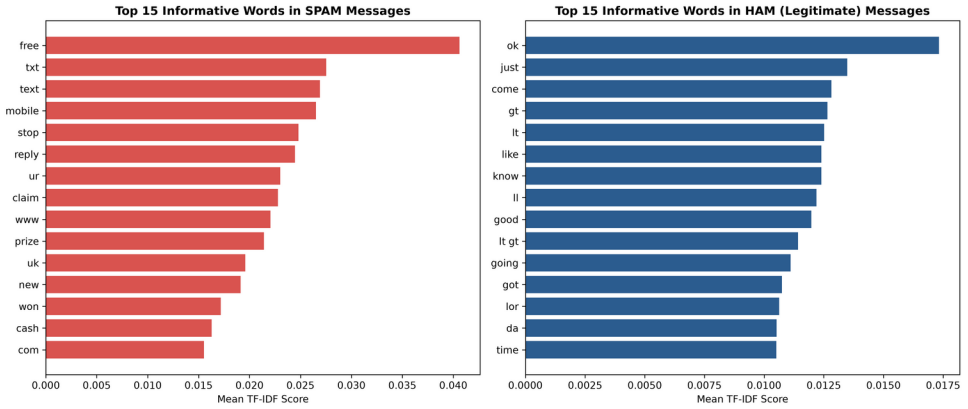
Logistic Regression (Primary SMOTE Model): Binary log-odds model using sigmoid activation: P(Y=1|X) = 1 / (1 + e^-(beta*X)). Optimized with L2 ridge regularization (C=2.0).

Impact of SMOTE on Logistic Regression: SMOTE synthesis expanded the minority spam decision region, dramatically cutting False Negatives from 28 to 13 and boosting Spam Recall from 78.63% to 90.08%.

Multinomial Naive Bayes Benchmark: Models feature probabilities using Bayes' rule with Laplace smoothing (alpha=0.2). Highly effective for sparse discrete word frequencies.

Why SMOTE Harms Naive Bayes: SMOTE generates continuous fractional features that violate discrete multinomial count assumptions and artificially imposes a 50/50 prior, causing test precision to drop. Standard MNB preserves 98.26% precision.

Pedagogical Insight: Demonstrates that oversampling techniques are not universally beneficial; geometric classifiers (LR) benefit from SMOTE, while generative Bayesian models perform best on natural frequencies.



Key Learning #3: Transformer Architectures (DistilBERT & RoBERTa)

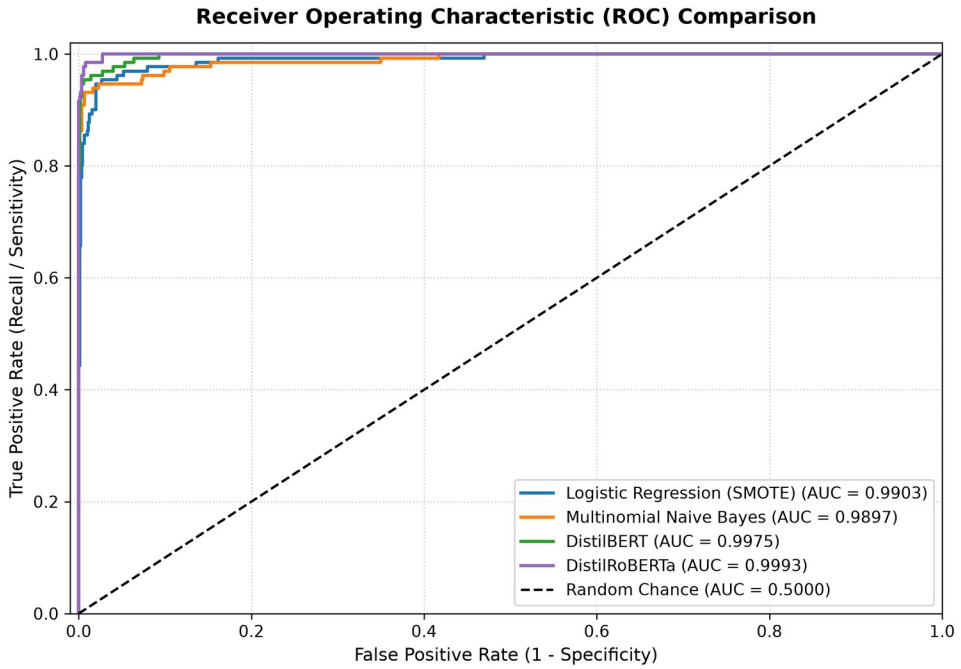
Beyond Bag-of-Words: Traditional TF-IDF ignores syntactic order and semantic nuance. Transformers use multi-head self-attention to capture bidirectional contextual relationships.

DistilBERT Architecture: 6-layer distilled transformer retaining 97% of BERT-base capability with 40% fewer parameters (66M) and 60% faster inference speed.

DistilRoBERTa Architecture: Distilled variant of RoBERTa (82M parameters), pre-trained on 160GB of text with byte-pair encoding and dynamic masking for superior semantic subtlety.

Fine-Tuning Methodology: Fine-tuned for binary sequence classification using Hugging Face Transformers and PyTorch with AdamW optimizer, warmup scheduling, and [CLS] pooling.

Colab GPU Training & Export: Reproducibly fine-tuned in Google Colab using free T4 GPU acceleration, exported to safetensors, and loaded for local CPU inference in Streamlit.



Empirical Evaluation: Test Set Benchmark (N = 1,034)

Test Split Integrity: All models evaluated on the identical, untouched held-out test split of 1,034 messages (903 Ham, 131 Spam).

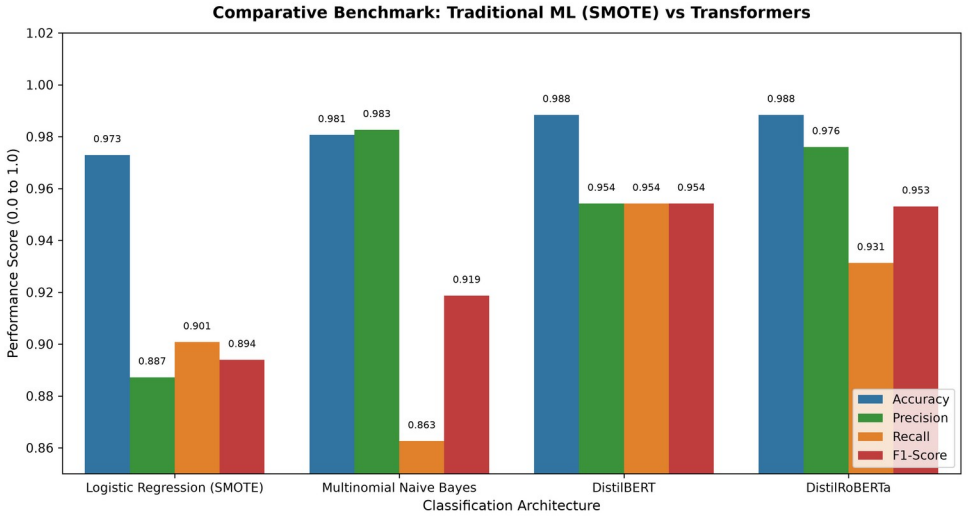
DistilBERT (Top Overall): 98.84% Accuracy | 95.42% Precision | 95.42% Recall | 95.42% F1 | 0.9975 ROC-AUC | 6 FP, 6 FN

DistilRoBERTa (Top Precision): 98.84% Accuracy | 97.60% Precision | 93.13% Recall | 95.31% F1 | 0.9993 ROC-AUC | 3 FP, 9 FN

Multinomial Naive Bayes: 98.07% Accuracy | 98.26% Precision | 86.26% Recall | 91.87% F1 | 0.9897 ROC-AUC | 2 FP, 18 FN

Logistic Regression (SMOTE): 97.29% Accuracy | 88.72% Precision | 90.08% Recall | 89.39% F1 | 0.9903 ROC-AUC | 15 FP, 13 FN

Key Benchmark Takeaway: Transformers deliver the highest balanced detection (F1 > 95.3%). However, traditional models achieve >97% accuracy with negligible latency.



Architectural Trade-Off Analysis: Traditional ML vs. Transformers

Performance & Accuracy: Transformers outperform traditional ML on F1-score (+3.5% to +6.0%) and ROC-AUC (0.999 vs 0.990) by deciphering conversational nuance, typos, and disguised links.

Training Speed Disparity: Traditional ML trains in 0.01 to 0.05 seconds. Transformers require ~45 minutes on GPU or multiple hours on CPU (~50,000x difference).

Inference Latency & Throughput: Traditional ML predicts in 0.0008 ms per message (~1.25 million msg/sec throughput). Transformers require ~70 ms per message (~14 msg/sec on CPU) — a 90,000x latency advantage for ML.

Hardware & Compute Footprint: Traditional models require <10 MB RAM and run on microcontrollers. Transformers require 260 MB disk storage, PyTorch runtimes, and hundreds of megabytes of RAM.

Operational Decision Framework: High-volume telecom SMS gateways should deploy Naive Bayes or Logistic Regression for real-time edge filtering, while high-value fraud detection and forensic auditing benefit from Transformers.

Interactive Streamlit Dashboard & Live Inferences

7-Section Streamlit Architecture: 1. Home 2. Dataset Overview 3. Data Exploration 4. Word Clouds 5. Model Performance 6. Model Comparison 7. Live Spam Prediction.

Live Inference Engine: Allows interactive testing across all 4 models with real-time confidence scoring and classification badges (SPAM vs. HAM).

Test Sample 1 (Spam): 'URGENT! You won £1,000 cash. Reply WIN to claim.' -> Predicted: SPAM (99.9% Confidence).

Test Sample 2 (Ham): 'Hi Sujal, are we still meeting in conference room B at 3:30 PM?' -> Predicted: HAM (99.8% Confidence).

Test Sample 3 (Disguised Spam): 'Free entry into the weekly £100 prize draw. Text WON to 85060' -> Predicted: SPAM (99.7% Confidence).

Ethical & Privacy Assurance: Zero persistent message logging, ephemeral memory inference, and bias auditing to protect user confidentiality.

